

INTERNSHIP REPORT-TASK 4

Title:Traffic Accident Data Analysis



Student Details

- **Name:** Keerthana Sangepu
- **Project Title:** Traffic Accident Data Analysis
- **Task:** 4 – Data Analysis & Visualization
- **Domain:** Data Science

1. Objective

The objective of this project is to analyze traffic accident data to identify meaningful patterns related to road safety. The study focuses on understanding how factors such as speeding, alcohol consumption, and state-wise conditions contribute to accident frequency. Additionally, the project aims to visualize accident hotspots and derive actionable insights to improve road safety.

2. Dataset Overview

The dataset used in this analysis contains state-wise traffic accident statistics, including:

- Total number of accidents per state
- Alcohol-related accident factor
- Speeding-related accident factor
- Insurance and claim-related variables
- State abbreviations for geographic comparison

This dataset helps in understanding how behavioral and environmental factors influence road accidents.

3. Data Preprocessing

The following data preparation steps were performed:

- Loaded dataset using Pandas
- Checked dataset structure and data types

- Handled missing and non-numeric values
- Selected only numeric columns for correlation analysis
- Organized data for visualization and insights

4. Exploratory Data Analysis (EDA)

The following analyses were performed:

◇ 4.1 Accident Distribution by State

Identified top accident-prone states using total accident counts.

◇ 4.2 Alcohol Influence Analysis

Examined the relationship between alcohol involvement and accident frequency.

◇ 4.3 Speeding Analysis

Analyzed how speeding contributes to increased accident rates.

◇ 4.4 Correlation Analysis

A heatmap was generated to understand relationships between numerical factors affecting accidents.

5. Data Visualizations

The following visualizations were created:

-  Bar chart of top 10 accident-prone states
-  Scatter plot: Alcohol vs Accidents
-  Scatter plot: Speeding vs Accidents
-  Correlation heatmap of all numerical features

These visualizations help in identifying patterns and risk factors effectively.

6. Key Insights

From the analysis, the following insights were observed:

- Certain states consistently show higher accident rates compared to others.
- Speeding is strongly correlated with increased accident frequency.
- Alcohol consumption is a significant contributing factor to accidents.

7. Conclusion

The analysis clearly shows that human behavioral factors such as speeding and alcohol consumption play a major role in traffic accidents. Identifying such patterns can help authorities implement stricter traffic regulations, awareness programs, and preventive measures to reduce accident rates.

8. Future Improvements

- Use larger real-world datasets (e.g., US Accidents dataset)
- Apply machine learning models for accident prediction
- Include geographic hotspot mapping using Folium or Plotly
- Analyze time-based patterns (day/night, seasonal trends)

Final Note

This project demonstrates practical skills in:

- Data cleaning
- Exploratory data analysis (EDA)
- Data visualization
- Insight generation